ModelIt!

Student Activity Pack

Why Do Some Seeds Travel Far?

The Amazing Ways Seeds Move to New Places

Grade 2 | NGSS 2-LS2-2 | Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How do you think seeds get to new places if plants cannot move?

2 Have you ever blown on a dandelion? What happened to the seeds?

3 Draw a seed that you think could travel very far — what features does it have?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Seed Shape | Wind Interaction | Travel Distance

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Fluffy Dandelion Seed

Seed Shape is fluffy, Wind is breezy

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Heavy Acorn

Seed Shape is heavy round, Wind is breezy

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Helicopter Maple Seed

Seed Shape is winged, Wind is breezy

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Animal Carriers — Animals like squirrels, birds, and dogs that carry seeds on their fur or in their bellies to new places

Water Flow — Rivers, streams, and rain that carry seeds from one place to another — coconuts float across oceans!

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do seeds travel to new places if plants cannot walk, run, or fly? Use evidence from your simulation to support your answers.

1 Why is it important for seeds to travel away from the parent plant?

2 Which seed design surprised you the most — the fluffy, winged, or heavy one?

8 STEM Challenge: Design a Super Seed

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YOUR ENGINEERING MISSION

Design and build a seed model that can travel as far as possible when dropped from the same height.

THE SCENARIO

A nature center wants to teach visitors about seed dispersal. Your team will design a model seed that travels the farthest when dropped from shoulder height. Use what you learned about seed shape and wind to make your design!

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE — PAPER, COTTON BALLS, TISSUE PAPER, PIPE CLEANERS?

2. WHAT SHAPE WILL YOUR SEED BE AND WHY DID YOU CHOOSE THAT SHAPE?

3. HOW WILL YOU TEST WHETHER YOUR SEED DESIGN TRAVELS FAR?

4. HOW CAN YOU MAKE YOUR SEED TRAVEL EVEN FARTHER AFTER THE FIRST TEST?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Botanists study plants and how they grow, spread, and survive. They work in forests, gardens, labs, and farms to help plants thrive. They earn $45,000–$85,000/year.